一，基本初等函数求导

(1)$\frac{dC}{dx}=0$(C 为常数)

(2)$\frac{d(x^{\mu})}{dx}=\mu x^{\mu-1}$

(3)$\frac{d(\sin x)}{dx}=\cos x$

(4)$\frac{d(\cos x)}{dx}=-\sin x$

(5)$\frac{d(\tan x)}{dx}=\sec^2 x$

(6)$\frac{d\cot x}{dx}=-\csc^2 x$

(7) $\frac{d(\sec x)}{dx}=\sec x\tan x$

(8) $\frac{d(\csc x)}{dx}=-\csc x\cot x$

(9)$\frac{d\,a^x}{dx}=a^x\ln x$

(10) $\frac{d\,e^x}{dx}=e^x$

(11)$\frac{d\,log_a x}{d\,x}=\frac{1}{x\ln a}$

(12) $\frac{d\,lnx}{d\,x}=\frac{1}{x}$

(13) $\frac{d\arcsin x}{d\,x}=\frac{1}{\sqrt{1-x^2}}$

(14) $\frac{d\arccos x}{d\,x}=-\frac{1}{\sqrt{1-x^2}}$

(15) $\frac{d\arctan x}{d\,x}=\frac{1}{1+x^2}$

(16) $\frac{d\arctan x}{d\,x}=-\frac{1}{1+x^2}$

二，求下列函数的微分

(1) $y=\frac{1}{x}+2\sqrt{x}$

解：$dy=\left(-\frac{1}{x^2}+\frac{1}{\sqrt{x}}\right)dx.$

(2) $y=x\sin 2x$

解：$dy=(\sin 2x+x\cos 2x\cdot 2)dx$

$=(\sin 2x+2x\cos 2x)dx.$

(3) $y=\frac{x}{\sqrt{x^2+1}}$

解：$dy=\frac{\sqrt{x^2+1}-x\frac{x}{\sqrt{x^2+1}}}{(\sqrt{x^2+1})^2}dx$

$=\frac{dx}{(x^2+1)^{\frac{3}{2}}}$

(4) $y=\mathrm{In}^2(1-x)$

解：$dy=2\mathrm{In}(1-x)\cdot\frac{-1}{1-x}dx$

$=\frac{2}{x-1}\mathrm{In}(1-x)dx.$

(5) $y=x^2e^{2x}$

解：$dy=(2xe^{2x}+x^2e^{2x}\cdot 2)dx$

$=2x(1+x)e^{2x}dx.$

(6) $y=e^{-x}\cos(3-x)$

解：$dy=[-e^{-x}\cos(3-x)+e^{-x}\sin(3-x)]dx$

$=e^{-x}[\sin(3-x)-\cos(3-x)]dx.$

(7) $y=\arcsin\sqrt{1-x^2}$

解：$dy=\left[\frac{1}{\sqrt{1-(\sqrt{1-x^2})^2}}\cdot\frac{-2x}{2\sqrt{1-x^2}}\right]dx$

$=-\frac{x}{|x|}\cdot\frac{dx}{\sqrt{1-x^2}}$

$=\begin{cases}\frac{dx}{\sqrt{1-x^2}},-1<x<0,\\ \frac{-dx}{\sqrt{1-x^2}},0<x<1.\end{cases}$

(8) $y=\tan^2(1+2x^2)$

解：$dy=[2\tan(1+2x^2)\cdot sec^2(1+2x^2)\cdot 4x]dx$

$=8x\tan(1+2x^2)\sec^2(1+2x^2)\,dx.$

(9) $y=arctan\frac{1-x^2}{1+x^2}$

解：$dy=\frac{1}{1+\left(\frac{1-x^2}{1+x^2}\right)^2}\cdot\frac{(-2x)(1+x^2)-(1-x^2)\cdot 2x}{(1+x^2)^2}dx$

$=-\frac{2x}{1+x^4}dx.$

(10) $s=Asin(\omega t+\varphi)$ $(A,\omega,\varphi$是常数)

解：$ds=[Acos(\omega t+\varphi)\cdot\omega]dt$

$=A\omega cos(\omega t+\varphi)dt.$

(11)$y=x\arcsin x+\sin(tanx)$

解：$dy=(\arcsin x+\frac{x}{\sqrt{1-x^2}}+\cos(\tan x)\sec^2 x)dx$

(12)$y=\ln\sin x$

解：$dy=\frac{1}{\sin x}d(\sin x)$

$=\frac{1}{\sin x}d(\sin x)'$

$=\frac{1}{\sin x}\cos x dx$

$=\cot x dx$

(13)$y=\ln(x+a)$

$$dy = \frac{d(x+a)}{x+a}$$

$$= \frac{dx}{x+a}$$

(14) $y = x^2 e^x$

$$dy = (2x \cdot e^x + x^2 \cdot e^x)dx$$

$$= x(2+x)e^x dx$$

(15) $y = \mathrm{x}^2 \ln x \cos x$

$$dy = (2xln\ x\ cos\ x + x\ cosx - x^2 lnx\ sinx)dx$$

(16) $y = \sqrt{x^2 - 1}$

$$dy = \frac{1}{2\sqrt{x^2-1}} \cdot 2xdx$$

$$= \frac{xdx}{\sqrt{x^2-1}}$$

(17) $y = x^2 e^{-ax^2}$

$$dy = \left[2x \cdot e^{-ax^2} + x^2 \cdot e^{-ax^2} \cdot (-2ax)\right]dx$$

$$= 2x(1 - ax^2)e^{-ax^2}dx$$

(18) $y = xe^{-ax^2}$

$$dy = \left[1 \cdot e^{-ax^2} + x \cdot e^{-ax^2} \cdot (-2ax)\right]dx$$

$$= (1 - 2ax^2)e^{-ax^2}dx$$

(19) $y = x^2 e^{-ax}$

$$dy = [2x \cdot e^{-ax} + x^2 \cdot e^{-ax} \cdot (-a)]dx$$

$$= x(2 - ax)e^{-ax}dx$$

(20) $y = xtanx$

$$dy = (tanx + x \cdot sec^2 x)dx$$

(21) $y = 5 + 3x - 4x^3$

$$dy = (3 - 12x^2)dx$$

(22) $y = 3x^4 - 2x^2 + 8$

$$dy = (12x^3 - 4x)dx$$

(23) $y = 2x^3 - 6x^2 + 7x$

$$dy = (6x^2 - 12x + 7)dx$$

(24) $y = e^{1-3x}cosx$

$dy = [e^{1-3x} \cdot (-3) \cdot cosx + e^{1-3x} \cdot (-sinx)]dx$

$= -e^{1-3x} \cdot (3cosx + sinx)dx$

(25) $y = \ln(1 + e^{x^2})$

$dy = \frac{1}{1+e^{x^2}} \cdot \frac{d\, e^{x^2}}{d\, x}\, dx$

$= \frac{1}{1+e^{x^2}} \cdot e^{x^2} \cdot \frac{d\, x^2}{d\, x}\, dx$

$= \frac{1}{1+e^{x^2}} \cdot e^{x^2} \cdot 2xdx$

$= \frac{2xe^{x^2}}{1+e^{x^2}}dx$

(26) $y = (x-a)(x-b)(x-c)$

$dy = \big((x-b)(x-c) + (x-a)(x-c) + (x-a)(x-b)\big)dx$

(27) $y = x\ln x - \frac{1}{2}x^2 + x$

$dy = \left(\ln x + \frac{x}{x} - x + 1\right) dx$

$= (\ln x - x + 2)dx$

(28) $y = x^3 + 3x^2 + 3x - 9$

$dy = (3x^2 + 6x + 3)dx$

$= 3(x+1)^2 dx$

(29) $y = 5x^3 - 2^x + 3e^x$

$dy = (15x^2 - 2^x ln2 + 3e^x)dx$

(30) $y = 2\tan x + \sec x - 1$

$dy = secx(2secx + tanx)dx$

(31) $y = x^2 lnx$

$dy = \left(2x \ln x + x^2 \frac{1}{x}\right) dx$

$= x(2\ln x + 1)dx$

(32) $y = \frac{lnx}{x}$

$dy = \frac{\frac{1}{x}\, x - 1 \ln x}{x^2} dx$

$$= \frac{1 - \ln x}{x^2} \mathrm{d}x$$

(33) $\mathrm{y} = \frac{e^x}{x^2} + ln3$

$$\mathrm{dy} = \frac{e^x\, x^2 + 2\, x\, e^x}{x^4} dx$$

$$= \frac{e^x(x-2)}{x^3} dx$$

(34) $\mathrm{y} = \frac{1+\sin t}{1+\cos t}$

$$\mathrm{dy} = \frac{\cos t(1+\cos t) - (-\sin t)(1+\sin t)}{(1+\cos t)^2}\, dt$$

$$= \frac{\cos t + \cos^2 t + \sin t + \sin^2 t}{(1+\cos t)^2}\, dt$$

$$= \frac{1 + \sin t + \cos t}{(1+\cos t)^2}\, dt$$